

F1 DRIVERS: USE OF MACHINE LEARNING TECHNIQUES TO PREDICT WHETHER A DRIVER WILL FINISH THE RACE OR NOT

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Abstract

In the high-stakes world of Formula 1 (F1), every second counts and so does every decision. In this project, the aim is to apply machine learning techniques to the F1 race data to predict whether a driver will finish a race or not. The algorithm is programmed to be applied after the qualifying and clearly before the race. The motivation behind this analysis is to identify patterns and factors that contribute to a driver's ability to complete a race, which can have important implications for constructors, analysts and fans. The objective is to build a classification model that, given the available data of a race (such as driver statistics, team information, historical performances of the drivers and of the teams, weather conditions and track characteristics), can estimate the probability of a driver finishing the race. It's important to note that the only information not available the day before the race concerns the weather conditions. In this case, meteorologists' forecasts are relied upon, forecasts which have become increasingly reliable and accurate in recent years. The entire machine learning analysis is performed using KNIME.

1 Introduction

This project began with a question in mind: *Will a driver make it to the finish line?*. Finishing a race might seem obvious or expected, but it is far from guaranteed. A wide range of elements can influence the outcome: driver experience and risk profile, mechanical reliability of the vehicle, and the unpredictable conditions of a race day. The goal is to understand which of these factors matter most and, more importantly, to develop a predictive model that can estimate, before the race begins, whether a driver will finish or not. Data from multiple sources is collected and integrated, building a comprehensive dataset in seven seasons of F1. For each race, a rich set of variables are considered: driver characteristics (age, number of races completed, historical incident rates, etc.), vehicle performance metrics (including

mechanical failures and average qualifying speed), as well as circuit complexity and weather conditions on race day. Using these data, various classification models are trained and tested in order to tackle two challenges: first, predict the likelihood that a driver finishes a race; second, understand which features among all those considered most strongly influence that outcome. The models are trained on six seasons of historical data and tested on the seventh, simulating a realistic forecasting scenario. In the end, the most accurate and robust model is selected, one capable of offering valuable insights into driver reliability and helping teams, analysts, or fans anticipate what might unfold even before the lights go out.

2 Dataset

The dataset [4] consists of all information on F1 races, drivers, constructors, qualifying, circuits, lap times, and championships from 2018 to the latest 2024 season. It is important to note that over the years there have been various team changes in Formula 1. However, between 2018 and 2024, all changes involved only the team names, while the staff and cars remained the same. Therefore, in the dataset, the most recent team name (chronologically) was retained and applied consistently across all seasons. Furthermore, the dataset was integrated with weather conditions and circuit characteristics loaded using APIs [3]. The data is made up of 2979 observations and 43 variables. , all the variables and their corresponding explanations can be seen at the following link [1]. The following is a list of the most important variables.

Race, driver and constructor information

- raceId: unique race identifier
- circuitId: unique circuit identifier
- location: circuit location
- date: date of the race
- driver_name: driver's name
- age: driver's age
- experience: number of races ran
- constructor_name: constructor name

Race performance

- grid: position of the starting grid
- prevraces_positiondiff: median of position differences (grid and final position) over the last five races
- q1_gap: best lap of the driver in Q1 session seen as the gap from the best one
- position_driver_championship: championship position of the driver before the race
- position_constructors_championship: championship position of the team before the race

- constructor_qtime_gap: average of the 2 fastest laps in the qualifying session for each team seen as the gap from the best average time

Incidents

- is_crash: 1 if the driver retired due to crash, 0 if not
- mechanical_problems: 1 if the driver retired due to mechanical issues, 0 if not.
- laplace_crash_in_circuit: Laplace-adjusted crash rate for the driver in this circuit, computed as $(\text{crashes} + 1) / (\text{races} + 2)$.
- recent%crashes: percentage of races not finished caused by crash in the previous 5 races
- recent_laplace_mechproblems: probability of experiencing mechanical issues in the race, calculated as the average Laplace estimate over the last 5 races ($k=1$)
- historical_%crashes_circuit: historical percentage of races ended due to a crash on that circuit

Weather and circuit informations

- n_corners: number of circuit's corners
- air_temperature: average air temperature
- track_temperature: average track temperature
- pressure: atmospheric pressure
- %rainfall: percentage of wet laps
- atmospheric_conditions: wet/dry race

Outcome

- retirement_target: 0 if the driver finished the race, 1 if he did not

3 Explorative Data Analysis (EDA) and Preprocessing

3.1 EDA

This section provides an initial analysis of the dataset. The statistical properties of the variables are examined, and the most relevant information for visualization is identified in relation to the objectives of the study. The structure of the dataset is first explored, revealing that all variables are numeric except for atmospheric_conditions, which is categorical.

Basic descriptive statistics were then computed using the Statistics node, including the number of missing values for each variable (Table 1).

Feature	Missing %
totcrashes_in_circuit	19.16
totraces_in_circuit	19.16
%crash_in_circuit	19.16
laplace_crash_in_circuit	19.16
historical_%crashes_circuit	6.04
historical_%mechproblems_circuit	6.04
recent%crashes	5.10
recent_laplace_crashes	5.10
recent_laplace_mechprob	4.69
q1_gap	1.00
prevraces_positiondiff	0.63

Table 1: Percentage of missing values by feature.

This overview facilitates a preliminary assessment of data quality and informs subsequent preprocessing and analysis steps.

The analysis proceeds by investigating the relationship between weather conditions and the likelihood of driver incidents. To visualize this relationship, a bar chart is employed to display the distribution of incident probabilities across different weather conditions (wet or dry) and types of incidents.

	crashes (%)	mech_issues (%)
Dry	6.96	7.65
Wet	13.08	6.78

Table 2: Probability of accident for weather conditions and type.

The values of Table 2 indicates that there are more retirements for crashes in wet conditions, but more retirements for mechanical issues in dry conditions.

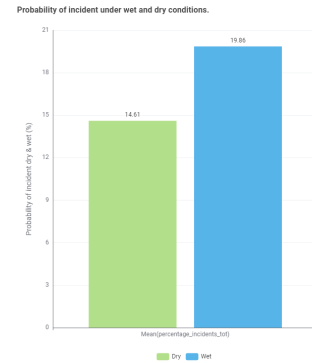


Figure 1: Probability of retirement for weather conditions.

The retirement rate in wet conditions is 19,86%, meaning that about 20% of drivers fail to finish the race when it's wet, compared to a 14,61% retirement rate in dry conditions.

The number of turns for each circuit is shown (Figure 2), providing an overview of the track characteristics, and the use of a histogram to display the distribution of the number of turns across circuits, helping to identify trends in circuit design.

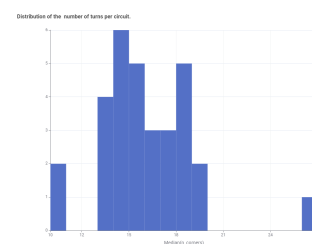


Figure 2: Number of turns per circuit

Finally, the distribution is examined of the retirement_target (Figure 3) and performed the same analysis on the subset of data containing missing values. The comparison shows no significant difference between the two cases.



Figure 3: Comparison between finished race and not finished.

Variable 1	Variable 2	<i>r</i>
age	experience	0.907
age	total_races	0.701
experience	total_races	0.763
alt	atm_pressure	-0.995
rainfall	Dry	-0.875
rainfall	Wet	0.875
pos_driver	pos_constructor	0.893
lap_crash_circ	%crash_circ	0.712
recent%_crashes	recent_lapcrashes...	0.852

Table 3: Correlation between independent variables.

3.2 Preprocessing

Prior to data preprocessing, rows corresponding to the first race of each year are observed to contain a higher proportion of missing values compared to other rows. Consequently, these rows are removed from the dataset. The initial preprocessing step involves imputing missing values in the training set. For variables exhibiting a higher percentage of missing data, Random Forest Regression is employed to predict missing values based on relationships and patterns identified among other variables. For variables with a lower percentage of missing values (such as *q1_gap* and *prevraces_positiondiff*), median imputation is applied. The same procedure is performed on the test set, utilizing the trained imputation model. Following imputation, variance checks confirm that the variance of the imputed variables remains approximately equal to that of the original data, thereby validating the imputation process.

In the second stage, an assessment of multicollinearity among the input variables is conducted, retaining only the most informative and representative features to improve model interpretability and performance. The same procedure is applied to both the training and test sets to maintain consistency. Table ?? reports pairs of variables with correlation coefficients greater than 0.7 or less than -0.7, with abbreviated names.

Additionally, a variance analysis is applied to eliminate features with extremely low variability, which are likely to contribute minimal predictive value and may introduce noise into the model. Subsequently a feature selection process that uses a Random Forest model identifies and retain only the most relevant variables. Features with an importance index below the threshold (greater than or equal to 30) are discarded, ensuring that only those with a strong predictive contribution are kept for downstream modeling. This feature selection step is likewise applied to both the training and test sets. These operations improve model efficiency and reduce the risk of overfitting by eliminating noise and redundancy in the dataset. In the final stage of the preprocessing pipeline, class imbalance is addressed through a hybrid approach that combines SMOTE-based oversampling of the minority class (1) with controlled undersampling of the majority class (0). This balancing procedure is applied exclusively to the training set. This strategy allows to achieve a balanced dataset while minimizing the risk of overfitting due to excessive synthetic sample generation, ultimately preserving the integrity and representativeness of the original data. Below the pie chart updated with the new distributions of 1 and 0:

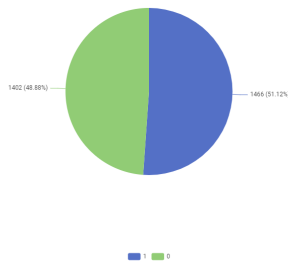


Figure 4: Classes balanced.

4 Classification Models and Performance Metrics

In this section, the classification models applied to the binary target variable `retirement_target` is described, which identifies whether a driver will finish a race (0) or retire due to an incident or car failure (1). The goal is to create a predictive system using only available data before the race.

4.1 Models and Configuration

A wide range of classification algorithms are tested in KNIME, covering linear and non-linear, generative and discriminative approaches. This diversity facilitates comparison among models of varying complexity and interpretability. Below is a summary of the models used and the main configuration and optimization strategies.

- **Naive Bayes (NB)**: is simple probabilistic classifier assuming conditional independence between features. It served as a baseline model. Laplace smoothing is enabled to handle zero probabilities. Due to its strong assumptions, it tends to underfit on more complex patterns.
- **Logistic Regression (LR)**: a linear model with L2 regularization, optimized for the regularization strength λ , convergence criteria, and maximum iterations. The model has been standardized prior to training. It offered interpretable results, but struggled to model non-linear relationships.
- **k-Nearest Neighbors (KNN)**: a nonparametric model sensitive to feature scaling. The

number of neighbors k are tuned (odd values between 3 and 15). A small k causes overfitting; a large k causes underfitting. Euclidean distance is used.

- **Support Vector Machine (SVM)**: is implemented with an RBF kernel and optimized for C and γ . Probability estimates were enabled using the Platt scaling. The model benefitted from class weighting to address imbalance and required careful tuning to avoid overfitting.
- **Decision Tree (DT)**: designed with maximum depth and minimum samples per leaf to prevent overfitting. The Gini index was used as the default split criterion, although entropy was also tested. Without pruning, the model tended to overfit; regularization parameters helped to improve generalization.
- **Random Forest (RF)**: an ensemble of Decision Trees, generally robust to overfitting. The key hyperparameters included the number of trees, the maximum depth, and the minimum leaf size. The model performed well, especially when using cost-sensitive learning and feature subsampling.
- **Gradient Boosting (GB)**: Boosted trees trained sequentially with a low learning rate and shallow trees. Adapted to learning rate, number of estimators, and tree depth. The model is highly performant, but sensitive to overfitting without early stopping or regularization.
- **Multi-Layer Perceptron (MLP)**: a feedforward neural network using ReLU activation and dropout regularization. The tuning focused on the number of neurons (e.g. 10–20), the learning rate and the number of epochs. Regularization (L2) and dropout (20–50%) were essential to avoid overfitting the relatively small dataset.

4.2 Validation Strategy and Metrics

All models are trained on six seasons of race data and tested on the seventh, simulating a real-world forecasting scenario. During training, k -fold stratified cross-validation (given the few rows, $k = 5$ was

chosen) was used for reliable performance estimation.

The following metrics are used to evaluate the models:

- **Area Under the ROC Curve (AUC):** measures the ability of the model to distinguish between the two classes, independent of the classification threshold.
- **Precision:** Precision determines the fraction of records that actually turns out to be positive in the group that the classification model has declared as positive. It is calculated using the formula:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (1)$$

- **Recall:** Recall measures the fraction of positive records correctly predicted by the classification model. It is calculated using the formula:

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (2)$$

4.3 Performance Summary

Table 4 summarizes the results obtained from cross-validation for all models tested. The main evaluation metrics considered were AUC, Precision and Recall.

Model	AUC	Class	Precision	Recall
NB	0.593	0	0.507	0.577
		1	0.644	0.577
LR	0.565	0	0.505	0.562
		1	0.624	0.569
KNN	0.813	0	0.42	0.795
		1	0.896	0.618
SVM	0.823	0	0.999	0.651
		1	0.488	0.997
DT	0.776	0	0.728	0.697
		1	0.721	0.751
RF	0.738	0	0.63	0.812
		1	0.861	0.709
GB	0.725	0	0.8	0.786
		1	0.792	0.806
MLP	0.818	0	0.716	0.804
		1	0.833	0.754

Table 4: Performance Metrics by Model (Cross-Validation Averages)

Generally, all models appear to have quite good discrimination ability, except NB (AUC = 59.3%) and LR (AUC = 56.5%) which, as a consequence, have lower Precision and Recall scores than the other models. From this first training phase, KNN, SVM and MLP can correctly predict the most number of drivers' retirements. Special attention is given to SVM, as it demonstrates a tendency to misclassify the majority of finishers as retirements. (Class 0: Precision = 99.9%), but still misses some real finishers (Class 0: Recall = 65.1%) and was able to predict all retirements (Class 1: Recall = 99.7%) but overestimating the number (Class 1: Precision = 48.8%). In general, SVM seems to be the best model, although it tends to overestimate the number of accidents.

4.4 Predictions Analysis

After finding the best hyperparameters combination for each model, predictions over the seventh and last season of the dataset was made. Then a ROC curves was plotted to visually compare the model discrimination performance models. As shown in Figure 5.

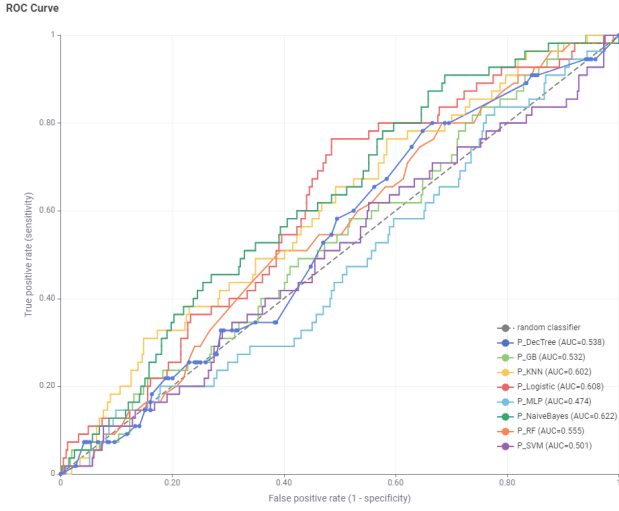


Figure 5: ROC Curves

Overall, the ROC curves do not show a model that performs better than others, but some "simpler" models that perform quite better than the others can be identified. NB (AUC = 62.2%), LR (AUC = 60.8%) and KNN (AUC = 60.2%) stood out as the most reliable classifiers. More "complex" models did not perform as well as in the training section; in fact, it was unable to find a model with an AUC greater than 56.0%, having the MLP model which falls under the random classification threshold. These results highlight a serious problem of overfitting the training data. Only NB and LR have shown improvements, thus the ability to generalize to the test data. This implies that the focus moves to these two models, plus the KNN that still maintains decent predictive ability.

Model	AUC	Class	Precision	Recall
NB	0.622	0	0.51	0.912
		1	0.636	0.15
LR	0.608	0	0.5	0.94
		1	0.764	0.172
KNN	0.602	0	0.465	0.913
		1	0.673	0.146
SVM	0.501	0	0.878	0.819
		1	0.164	0.11
DT	0.53.8	0	0.881	0.75
		1	0.122	0.255
RF	0.555	0	0.772	0.884
		1	0.255	0.132
GB	0.532	0	0.859	0.885
		1	0.182	0.149
MLP	0.474	0	0.715	0.873
		1	0.236	0.102

Table 5: Performance Metrics on the Test Set

Generally, all models cannot predict more than the 20% of the retirements, only the DT predict more (Class 1: Recall = 25.5%). On the other hand, all models predict almost the totality of the finishers correctly, touching the peaks with the models with the best AUC, with a Recall for Class 0 equal to 91.2%, 94% and 91.3%, respecting NB, LR and KNN, this means that they make prediction errors even for the finishers. The Precision follows a similar trend; the three models with the greatest AUCs predict a few retirements that turns out to be true, retirements are predicted even better than the finishers. The other models make accurate predictions over the finishers, because they tend to predict every driver as finisher, while for retirements they have not the same ability compared to the other three models, RF has the best Precision with a score of 25.5%.

4.5 Lift Analysis

The Lift Curve (LC) is used to determine the best model among the three chosen models. LCs show the Lift Factor (LF) at various percentiles of the predicted probability scores. The LF is calculated using the following formula:

$$LF = \frac{Performance(M)}{Performance(ZeroRule)} \quad (3)$$

where:

- $Performance(M)$: the performance of the chosen model
- $Performance(ZeroRule)$: the performance of the casual sampling

The following LCs are generated to choose the best model.

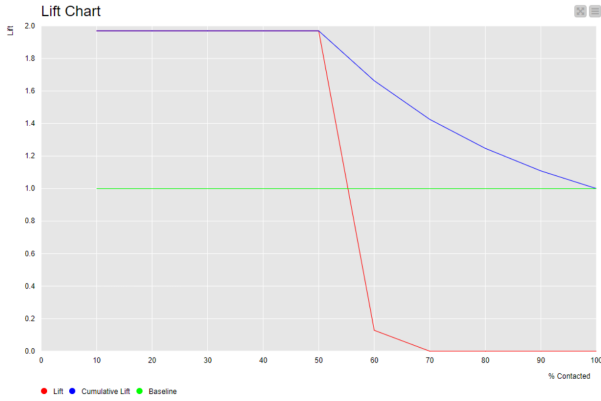


Figure 6: Lift Naive Bayes

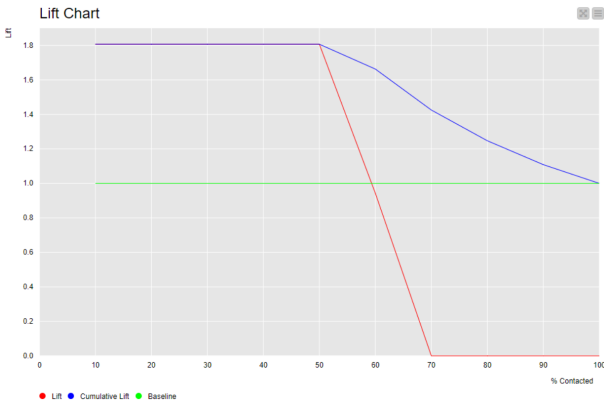


Figure 7: Lift Logistic Regression

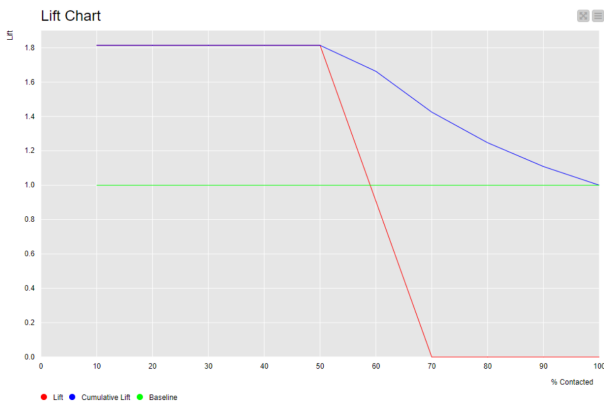


Figure 8: Lift KNN

As expected, none of the chosen models maintains a particularly high lift line compared to the oth-

ers, but they have similar behaviors. So, we face a dual choice:

- NB: has a higher lift at the 50% threshold, it indicates that it does a better job early on, but comes with a quick drop-off after 55%
- LR and KNN: they maintain a relatively strong performance at higher percentiles, since it crosses the random line later (60%) and has a more gradual decline

Since the goal of the work is to prevent the retirement of drivers, we choose the model with the highest lift (NB = 2), to better identify the drivers who are more likely to retire.

5 Conclusions and Future Developments

Despite several attempts, it is difficult to find a good trade-off between a good overall balance of the data and a sufficient discrimination ability of the models. All models exhibit difficulty in performing effectively on the unbalanced dataset, while a few demonstrate a decent performance with balanced data. Based on AUC, Precision and Recall NB, KNN and LR emerge as the most reliable classifiers. However, none of these three models significantly outperformed the others.

Among the three best models, NB stands out as the top performer in the investigation. It has a modest predictive ability, with an AUC of 62.2% on the test set. However, its main weakness is the poor recall for identifying retirements (Class 1: Recall = 15%), meaning that the model misses a significant number of actual retirements. On the positive side, the model maintains decent precision for both classes, suggesting that the few predicted retirements are mostly accurate (Class 0: Precision = 51%, Class 1: Precision = 63.6).

A key challenge in this analysis is that many features relevant to predicting race outcomes are qualitative or external to the dataset, such as car reliability or driver confidence on specific circuits. Consequently, the available data is insufficient to support reliable predictions of a driver's probability of completing a race.

In summary, predicting the drivers who are more likely to retire during a specific race is a challenging goal that faced multiple obstacles during the work:

- Lack of data: many data are not available before the 2018 season
- Incompleteness of data: the available data are not specific enough to make accurate predictions; more specific information would have been useful
- Lack of bibliography: online researches do not provide helpful information, since no prior documentation on this topic was found
- FIA Regulation [2]: FIA updates its regulation annually, bringing with it both small and big changes that modify the structure of the cars and, as a consequence, the behavior of the drivers, thus the causes of the retirements across the years
- Unpredictability: some causes of retirement are totally unpredictable (e.g., an animal that crosses the track)

Successing with this goal with high accuracy is almost impossible given too many factors external to the track and huge casualty component that embraces driver behavior in case

of collision or accident or mechanical failure due to human error or single component failure. The model we built surely can be improved with additional information on the cars, drivers' driving style, realibility history of the constructors, the features of the circuits and variables that represent the external factors.

References

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